DAY 27 | POST-READ

Agents, Agentic RAG and Memory: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
Agent loop	Goal, plan, act, observe, reflect, repeat until done or stopped.
Components	Model, instructions, tools, memory, guardrails, orchestration.
Agents SDK	Agent, @function_tool, Runner, handoffs, guardrails, tracing.
Hosted agents (NEW v4)	Provider-run loops with computer use, multi-agent and compaction.
CRAG	Retrieve, grade, correct (rewrite or search elsewhere), generate.
Multi-agent (NEW v4)	Supervisor and specialists only when roles differ; every hop costs tokens.
Background agents (NEW v4)	Checkpoint every step; expose status; set time, step and token limits.
Memory (NEW v4)	Short-term per thread; long-term per user in a store; extract, screen, expire, delete.
Memory poisoning (NEW v4)	OWASP ASI06. Allowlist memory kinds; memories are data, not instructions.

Part B: Going beyond today

1. Self-RAG and reflection

Beyond grading documents, agents can grade their own draft: "Is every claim supported by the evidence?" If not, revise. This costs extra calls but reduces unsupported answers.

2. Semantic memory search

Stores can index memories with embeddings, so the agent recalls the memories most related to the current question instead of all of them. This saves tokens as memories grow.

3. Managed memory services

Cloud platforms offer managed long-term memory, for example Vertex AI Agent Engine Memory Bank, which extracts and consolidates memories for you. Check data residency, retention and deletion controls before choosing one.

4. Agent evaluation

Evaluate agents on outcomes (task solved?), process (sensible tool use, steps taken) and safety (no forbidden actions, no poisoned memories). You will build trajectory evals on Day 31.

5. Cost control

Cap steps and tokens per task, use lighter models for grading and routing, cache tool results, and log cost per run and per hop.

Part C: Check yourself

- What makes an agent different from a single LLM call?
- What does the grading step in CRAG decide?
- Give two reasons not to split a task across several agents.
- What is the difference between the checkpointer and the store in LangGraph?
- How does memory poisoning work, and name two defences.

Part D: Practice ideas for tonight

- Add a reflection node to the CRAG graph that checks the answer against the evidence.
- Add a 'contact channel' memory and recall it in a second session.
- Write two more poisoning tests: one in English, one hidden in a long message.

Coming up on Day 28: agent protocols (MCP 2026-07-28 and A2A v1.0), a sandboxed data agent and a senior FDE interaction.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today